

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Artificial Intelligence- Driven X-Ray Analysis: Enhancing Image Quality and Diagnostic Accuracy

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Artificial Intelligence- Driven X-Ray Analysis: Enhancing Image Quality and Diagnostic Accuracy**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Monika Aggarwal**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Through medical imaging, it is possible to find and be diagnosed with diseases, especially for patients who have a respiratory condition like pneumonia. Early diagnosis is extremely important in improving the overall outcomes for a patient. There are many ways of doing medical imaging; however, of the different types of imaging techniques, chest x-rays are the most common imaging technique used because they are low cost, widely available, and easy to complete. One of the biggest problems with chest x-ray images is the difficulty in accurately interpreting those images; due to differences in image quality and contrast, as well as the number of subtle abnormalities, it is often very difficult to tell the difference between a normal and an abnormal image when viewed manually. To solve some of the limitations associated with manually identifying chest x-ray images, we propose using an artificial intelligence (AI)-based imaging system that improves the quality of chest x-ray images, while increasing the diagnostic accuracy of those efforts. For realizing these objectives through this research, some main objectives have been set. Firstly, there is an attempt to improve the quality of the image for the purpose of analyzing the data effectively. Then, there is an objective that seeks to develop a strong classification model with the help of deep learning approaches in order to detect any kind of anomalies. The proposed system integrates advanced techniques for preprocessing images to enhance their quality, along with an AI-based deep learning model using the ResNet-18 architecture, and transfer learning to create an accurate predictive model. The model was developed using a publicly available data set, and then validated using a separate data set, in order to determine if the model had the ability to generalize and would be robust to use in the future. The results from the experiments show that the proposed model is highly accurate and has a good recall rate when distinguishing normal cases from abnormal images, and is a potential method of assisting in the early-diagnosis of patients. Grad-CAM visualizations were also generated during the experiments and were used to assist with interpretability of the AI predictive model's results and decision-making process; which increase transparency and improve trust in the use of the model for clinical applications.

INTRODUCTION

Chest X-ray imaging is one of the most widely used medical imaging techniques for diagnosing respiratory and lung-related diseases such as pneumonia, tuberculosis, lung infections, and other chest abnormalities. It is commonly preferred in hospitals and healthcare centers because it is cost-effective, fast, and easily accessible. Early and accurate diagnosis of chest diseases is extremely important, as delayed detection can lead to severe health complications and increased mortality rates.

Despite its importance, manual interpretation of chest X-ray images remains a challenging task for radiologists and healthcare professionals. Variations in image quality, low contrast, overlapping anatomical structures, and subtle abnormalities often make diagnosis difficult. In addition, radiologists are required to examine a large number of X-ray images daily, which may result in fatigue, delayed diagnosis, and human errors. These challenges highlight the need for intelligent automated systems that can assist healthcare professionals in improving diagnostic accuracy and efficiency.

In recent years, Artificial Intelligence (AI) and Deep Learning techniques have shown remarkable success in the field of medical image analysis. Convolutional Neural Networks (CNNs), a class of deep learning models, are capable of automatically extracting important features from medical images and performing accurate classification tasks. Advanced architectures such as ResNet, VGG, and DenseNet have demonstrated significant improvements in disease detection using chest X-ray images. Among these, ResNet-18 has gained popularity due to its strong performance, reduced computational complexity, and ability to overcome vanishing gradient problems using residual connections.

This research focuses on developing an AI-driven chest X-ray analysis system that enhances image quality and improves diagnostic accuracy. The proposed system combines image preprocessing techniques such as Gaussian filtering, Contrast Limited Adaptive Histogram Equalization (CLAHE), and edge sharpening with a deep learning model based on ResNet-18 using transfer learning. These preprocessing techniques help improve image clarity and contrast, enabling the model to identify abnormalities more effectively.

The study also incorporates Grad-CAM (Gradient-weighted Class Activation Mapping), an

explainable AI technique that generates heatmaps to highlight the regions of the X-ray image responsible for model predictions. This improves transparency and helps radiologists understand the decision-making process of the AI system.

The primary objective of this research is to develop a reliable and efficient diagnostic support system that can assist healthcare professionals in the early detection of chest diseases. By integrating image enhancement, deep learning, and explainable AI techniques, the proposed system aims to improve classification accuracy, reduce diagnostic errors, and support better clinical decision-making in real-world healthcare environments.

METHODOLOGY

The methodology adopted for this research is analytical and implementation-oriented. The study focuses on developing an AI-based chest X-ray analysis system using image preprocessing, deep learning, and explainable AI techniques to improve disease detection accuracy.

1. Literature Review

A detailed review of research papers, journals, and case studies related to medical image analysis, chest X-ray classification, deep learning, and explainable AI was conducted. Existing techniques such as CNN, VGG, DenseNet, and ResNet architectures were studied to understand their advantages and limitations in medical diagnosis.

2. Dataset Collection

A publicly available chest X-ray dataset was collected containing normal and abnormal chest X-ray images. The dataset was analyzed and organized for training, validation, and testing purposes.

3. Image Preprocessing

To improve image quality and model performance, several preprocessing techniques were applied:

- Gaussian filtering for noise reduction
- CLAHE (Contrast Limited Adaptive Histogram Equalization) for contrast enhancement
- Edge sharpening for better feature visibility
- Image resizing and normalization for consistent model input

4. Model Development

A deep learning model based on ResNet-18 architecture was developed using transfer learning. The pre-trained model was fine-tuned to classify chest X-ray images into normal and abnormal categories.

5. Model Training and Evaluation

The model was trained using the processed dataset and evaluated using standard performance metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score

Training and validation loss curves were also analyzed to measure model performance and prevent overfitting.

6. Explainable AI Analysis

Grad-CAM (Gradient-weighted Class Activation Mapping) was used to generate heatmaps highlighting the important regions of chest X-ray images responsible for model predictions. This step improved the transparency and interpretability of the system.

7. Comparative Performance Analysis

The proposed ResNet-18 model was compared with other deep learning models such as CNN, VGG, and DenseNet based on classification accuracy, recall, precision, and overall performance. The comparison helped evaluate the effectiveness of the proposed approach.

TOOLS AND TECHNOLOGIES

This project utilizes various tools and technologies for chest X-ray image processing, deep learning model development, performance evaluation, and result visualization.

Programming Language

- **Python** – Used for image preprocessing, model training, evaluation, and implementation of deep learning algorithms.

Deep Learning Framework

- **PyTorch** – Used for developing and training the ResNet-18 deep learning model.

Image Processing Libraries

- **OpenCV** – Used for image enhancement techniques such as filtering, sharpening, and preprocessing.
- **NumPy** – Used for numerical computations and array operations.

Visualization and Analysis Tools

- **Matplotlib** – Used for plotting graphs such as accuracy curves, loss curves, ROC curves, and confusion matrices.
- **Scikit-learn** – Used for calculating performance metrics including accuracy, precision, recall, F1 score, and ROC-AUC score.

AI and Explainability Tools

- **ResNet-18** – Deep learning architecture used for chest X-ray classification.
- **Grad-CAM** – Used for generating heatmaps and visualizing important image regions responsible for model predictions.

IMPLEMENTATION

The implementation phase of this project focuses on the development of an AI-based chest X-ray analysis system using image preprocessing, deep learning, and explainable AI techniques for disease detection.

1. Image Acquisition and Preprocessing

Chest X-ray images were collected from a publicly available medical imaging dataset. Before training the model, preprocessing techniques were applied to improve image quality and enhance important features. The preprocessing stage included:

- Gaussian filtering for noise reduction
- CLAHE for contrast enhancement
- Edge sharpening for highlighting lung structures
- Image resizing and normalization for uniform input dimensions
- These techniques improved the visibility of abnormalities and helped the model learn more effectively.

2. Deep Learning Model Implementation

The ResNet-18 architecture was implemented using the PyTorch framework. Transfer learning was applied by utilizing pre-trained weights, which reduced training time and improved classification performance. The final fully connected layer of the model was modified for binary classification of chest X-ray images into normal and abnormal categories.

3. Model Training and Validation

The processed dataset was divided into training, validation, and testing sets. The model was trained using optimized parameters and data augmentation techniques such as image rotation and flipping to improve generalization. During training:

- Training loss and validation loss were monitored
- Accuracy curves were analyzed
- Overfitting was reduced using validation-based optimization

4. Performance Evaluation

After training, the model was evaluated using standard classification metrics, including:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score

Confusion matrices, ROC curves, and precision-recall curves were generated to analyze the effectiveness of the proposed system.

5. Explainable AI Implementation

Grad-CAM (Gradient-weighted Class Activation Mapping) was implemented to generate heatmaps highlighting the regions of chest X-ray images responsible for the model's predictions. This helped improve transparency and allowed better interpretation of the AI-based diagnosis system.

6. Comparative Analysis

The performance of the proposed ResNet-18 model was compared with other deep learning models such as CNN, VGG, and DenseNet. The comparison was performed based on classification accuracy, recall, precision, and overall reliability in detecting chest abnormalities.

RESULTS

The findings of this research demonstrate that the integration of image preprocessing techniques, deep learning, and explainable AI significantly improves the performance of chest X-ray classification systems. The proposed AI-based diagnostic model achieved high accuracy and reliability in detecting abnormalities from chest X-ray images.

The experimental results showed that the proposed ResNet-18 model achieved:

- **Accuracy:** 91.5%
- **Recall (Sensitivity):** 96.5%
- **F1 Score:** 92.6%

The high recall value indicates that the system is highly effective in identifying abnormal cases and minimizing false negative predictions, which is extremely important in medical diagnosis because missed disease cases may delay treatment and increase health risks.

The implementation of image preprocessing techniques such as Gaussian filtering, CLAHE, normalization, and edge sharpening improved the quality of chest X-ray images and enhanced the ability of the model to capture important features. As a result, the classification performance improved compared to traditional approaches.

The comparative analysis demonstrated that the proposed model outperformed several existing deep learning models such as CNN, VGG, and DenseNet in terms of overall classification performance and disease detection capability.

Another important finding of this research is the successful implementation of Grad-CAM visualization. The generated heatmaps clearly highlighted the regions of the chest X-ray images responsible for model predictions. This increased the transparency, interpretability, and trustworthiness of the AI system, making it more suitable for real-world clinical applications. Overall, the results confirm that combining image enhancement techniques with transfer learning and explainable AI can significantly improve chest X-ray analysis and support doctors in faster and more accurate medical diagnosis.

CONCLUSION AND FUTURE SCOPE

This research successfully demonstrates the potential of Artificial Intelligence in improving chest X-ray analysis and medical diagnosis. By combining image preprocessing techniques, deep learning models, and explainable AI methods, the proposed system achieved high accuracy and reliability in detecting abnormalities from chest X-ray images.

The study shows that techniques such as Gaussian filtering, CLAHE, normalization, and edge sharpening significantly enhance image quality and improve the model's ability to identify important medical features. The implementation of the ResNet-18 architecture with transfer learning further improved classification performance while reducing training complexity and computational effort.

One of the major strengths of the proposed system is the integration of Grad-CAM visualization, which increases the transparency and interpretability of the AI model. The generated heatmaps help radiologists and healthcare professionals understand the regions influencing the model's predictions, thereby improving trust and confidence in AI-assisted diagnosis.

Future Scope

- Development of more advanced hybrid deep learning architectures for multi-disease detection
- Integration of real-time AI-assisted diagnosis systems in hospitals and healthcare centers
- Use of larger and more diverse medical imaging datasets to improve model generalization
- Implementation of lightweight AI models for deployment on mobile and edge devices
- Enhancement of explainable AI techniques to improve transparency and clinical trust
- Integration of the proposed system with telemedicine platforms for remote healthcare support
- Expansion of research toward detecting multiple chest-related diseases simultaneously using a single AI framework

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